SOFTWARE ENGINEERING FOR HPC

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Exam rules

- Written exam (up to 16 points, minimum required 8 points). Software testing and analysis exercises. Exercises focusing on design aspects.
- Practical activities (up to 16 points, minimum required 8 points). A simple requirement analysis and software design assignment. Setup a tool pipeline automating software integration, deployment, some operation steps.

1 Introduction

Software engineering is a field of computer science dealing with software systems. Programming skills are not enough:

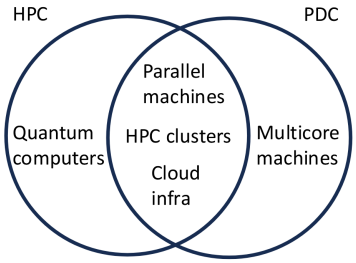
- A *programmer* develops a complete program, working individually (alone) on known specifications;
- A *software engineer*, instead, identifies requirements and develops specifications, designing a component to be combined with other components, developed, maintained, used by others. This component can become part of several systems. Most importantly, a software engineer works in a team.

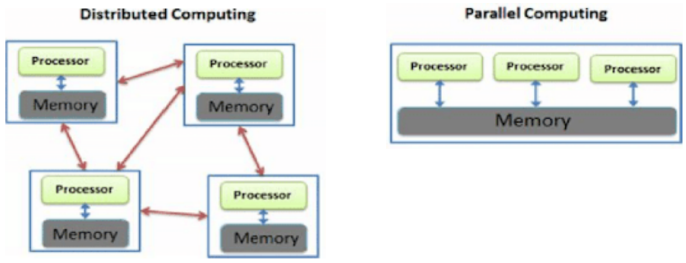
Initially, there was no reference model. As a reaction to the many problem, it was born the traditional *waterfall* model and, later, flexible processes: *iterative models*, *agile methods*, *DevOps*.

2 HPC software, relevant qualities and software engineering methods

High performance computing (HPC) is the platform technology concerned with programming for performance (speed), energy efficiency (doing more with less power), upscaling (handling larger problems), or high throughput (handle large volumes of data in near real-time).

Several years ago, HPC and *parallel and distributed computing* (PDC) were regarded as the same thing. Nowadays, they are distinct (but overlapping.



The main characteristics of PDC are *concurrency* (more than one active execution context), *parallelism* (execution of different tasks at the same time), and *distribution* (execution of different tasks on physically different interconnected computing nodes, without a

global clock). The following image shows the differences between distributed and parallel computing.